Multi-wavelength Object Classifier

λ ↑

Radio

IR

Optical

UV

X-ray

Give it a position: it gathers 29 archives,
sorts star · galaxy · QSO, and adds subclasses

Pipeline · tech stack summary v2.44.0 2026.11.01

At a Glance

Observations → evidence → probability → class. Every decision cites a reference and a measurement

29

Archives

Radio · IR · Optical · UV · X-ray · Spectroscopy · Literature

7

Stages

Every stage from collection to subclass writes a CSV

36 + 2

Rules + models

36 literature rules, LS-GBM and fusion GBM

99.16 %

Blind accuracy

24,636 spectroscopic truths never used in training (400 fields)

Coordinates → Collect → Associate → Integrate → Features → Evidence → **Fuse** → Subclass → App

One field (12′ radius, ~16k objects): collection ~5–8 min + analysis ~3 min

Workflow

Pipeline

`v2/pipeline.py` runs every stage in order and writes each stage's CSV and wall time

0
Collect
453 s

Get_data/controller.py

Cone search of 29 archives
6 in parallel

1
Associate
9 s

01_source_association

Group detections by
per-survey position errors

2
Integrate
13 s

02_integrate_objects

One object = one row
(~1,800 columns)

3
Features
18 s

03_extract_features

AB mag · color · extinction
Tractor fit quality

4
Evidence
16 s

04_build_evidence

36 literature rules +
LS-GBM · color kNN

5
Fuse
24 s

05_likelihood_vectors

p(star) · p(galaxy) · p(QSO)
priors · special rules

core decision

6–7
Subclass
42 s

Classifier/control.py

WD · giant · variable
AGN · SN …

+
Wrap-up
~40 s

host_groups · high_confidence

Host grouping
confidence flag · coverage warning

Times measured on field ngp_185m01 (15,820 objects), 2026-10-30. Collection time depends on archive load

29 Archives

Given a position and radius, every collector fetches the same field in parallel

Radio	IR	Optical	UV	X-ray
LoTSS	AllWISE	Gaia DR3	GALEX	Chandra
TGSS	2MASS	Gaia variability		XMM
NVSS		Legacy Surveys DR10		eROSITA
FIRST		Pan-STARRS1		
VLASS		SDSS DR18		
		ZTF periodic		



long λ — short λ

Spectroscopy 4

SDSS · DESI DR1 · DESI FastSpecFit · LAMOST

Catalog · Literature 8

SIMBAD · NED · ATNF · BZCAT · SGA-2020 · MYStIX · AGB · galaxy cluster

Keeping runs alive

6 in parallel

300 s per collector; a slow one is dropped alone

3–5 s spacing

A pause between requests (avoids archive blocking)

6 VizieR mirrors

No connection in 10 s or 5xx → next mirror

Gaia fallback

Slow ESA TAP → VizieR copy + fidelity table

90-day cache

Re-runs of a field skip the archives

Missing-catalog warning

Shown in the results and an app banner

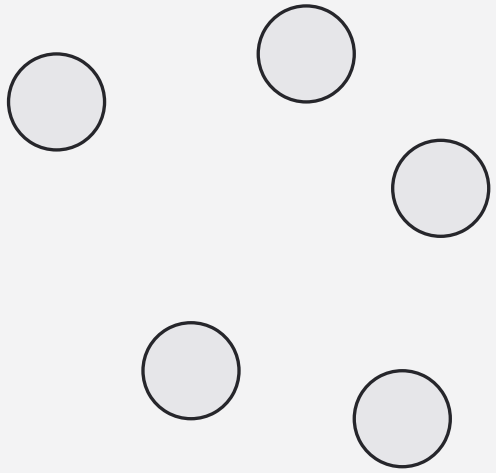
Detections → Objects

Detections from each catalog are merged into objects and turned into comparable numbers

26,687 detections



15,820 objects



Position error · proper motion give a same-object probability

Objects × features

g − r	**0.62**
z − W1	**1.85**
W2 − W3	**2.9**
parallax / σ	**0.4**
REX−PSF gain	**−0.3**
size significance	**1.6**

(example values)

1 Association

2 Integration

3 Features

All magnitudes in AB · Galactic extinction corrected · colors only at S/N ≥ 3

Building Evidence

One rule = one reference. The rule table stores the criterion, source and reliability; results are kept as an evidence list

Spectroscopy

strongest

Spectral class from SDSS · DESI · LAMOST

Astrometry

moving → star

Gaia parallax · proper motion significance
(Gaia DR3, Klioner+22)

Morphology

extended → galaxy

Legacy Surveys · PS1 · SDSS
extended / point-source flags

Color rules

SED shape

WISE AGN (Stern+12, Assef+18),
high-z QSO color

Features

new in 2.44.0

LS color · z−W1 locus · dchisq
· WISE color → learned weights

Literature labels

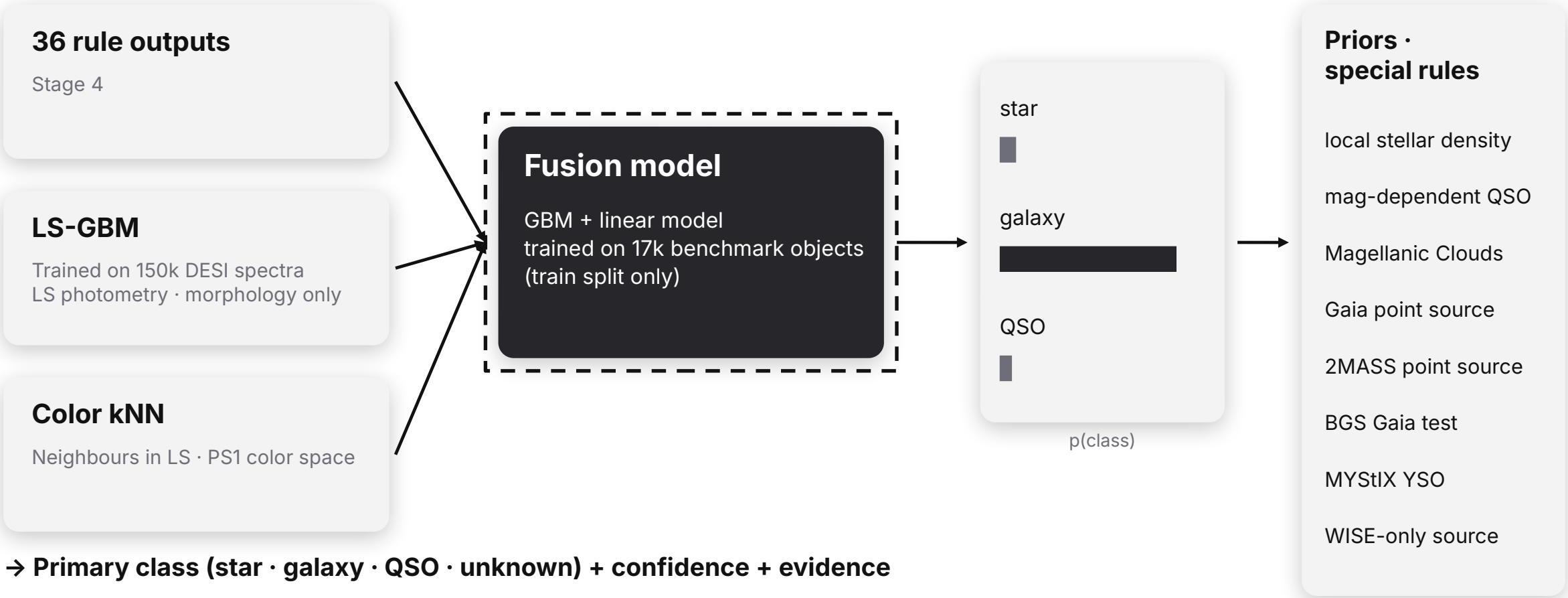
reference

SIMBAD · NED class
(excluded in blind validation)

Each rule emits { class, score, reliability, association reliability, source } → weighed together in Stage 5

Evidence → Probability

A trained model merges the evidence into one probability; literature priors adjust it last

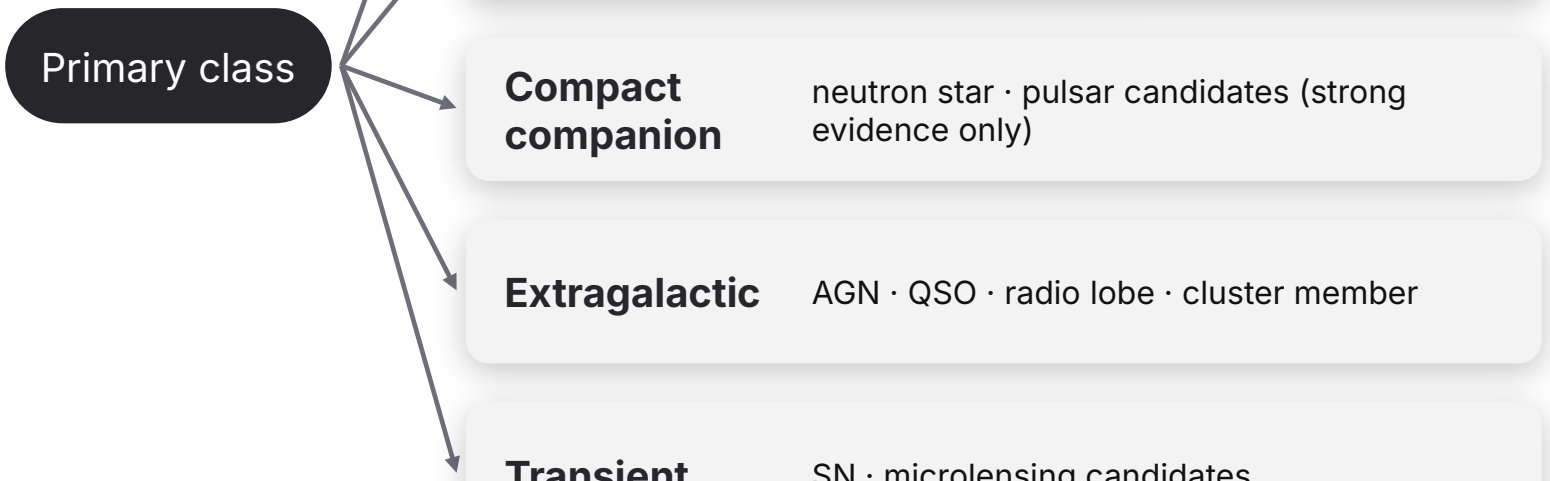


→ Primary class (star · galaxy · QSO · unknown) + confidence + evidence

Low confidence or conflicting evidence → left as unknown (no forced choice)

Subclassification

Independent axes are added on top of the primary class. An axis stays empty when the evidence is too weak



Physical	WD (DA/DB) · RGB · AGB (C-rich / O-rich)
Variability	RR Lyrae · Cepheid · LPV · eclipsing binary · Mira
Compact companion	neutron star · pulsar candidates (strong evidence only)
Extragalactic	AGN · QSO · radio lobe · cluster member
Transient	SN · microlensing candidates

Post-processing

Host grouping

Fragments inside a large galaxy are grouped under it (host_groups)

High-confidence flag

Spectroscopic confirmation or real-sky purity ≥ 98 % (gold ring in the app)

Missing-data warning

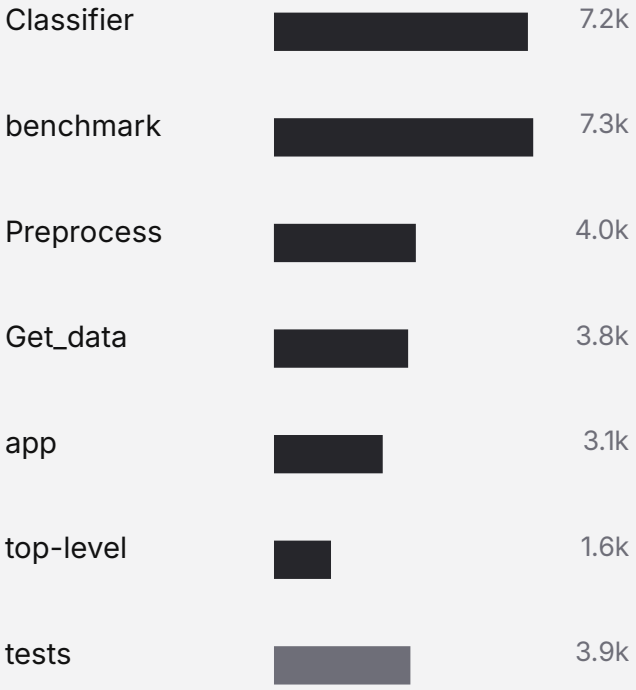
Fields missing a key archive are flagged in results and the app

Tech Stack

All Python. The app ships no ML library: trained trees are evaluated directly in numpy

App	**PySide6 (Qt) · PyInstaller** Windows · macOS · Linux builds (GitHub Release)
Models	**numpy inference · scikit-learn (training only)** Trees exported to npz keep the app small
Data	**pandas · numpy · scipy · astropy** Tables, neighbour search (cKDTree), coordinates · units
Archive access	**astroquery · pyvo (TAP) · requests** VizieR · ESA Gaia · NOIRLab Data Lab · MAST · CDS
Ops · QA	**GitHub Actions · pytest** Collection · validation · app builds as workflows, 141 tests

Code size



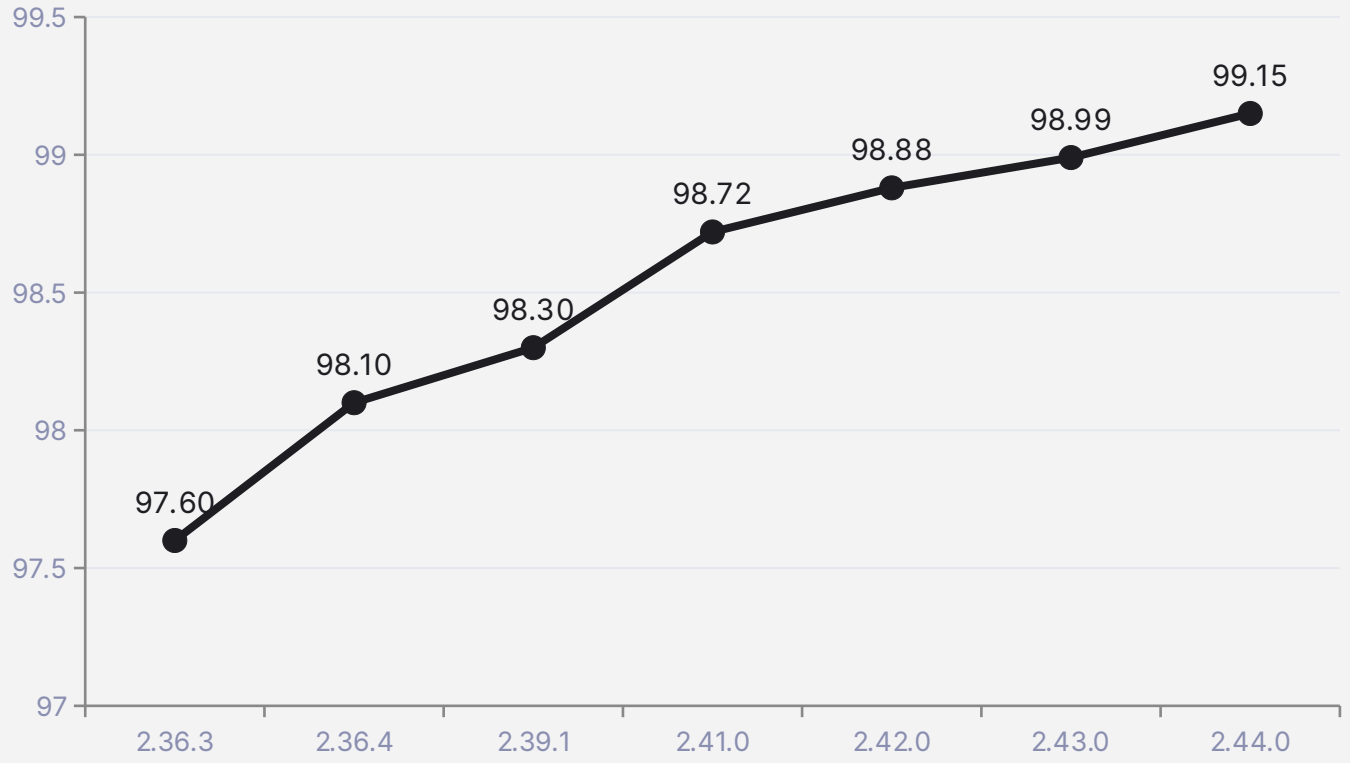
~31k lines of Python · 26 docs (references · work reports)

Accuracy

Validation

Scored against spectroscopic surveys never used in training
(LAMOST · 2QZ · 2dFGRS · 6dFGS · WiggleZ · APOGEE …)

Independent-field blind accuracy (set A)



99.16 %

400 independent fields
24,636 truth objects

99.7 – 99.8 %

High-confidence results
(~23k objects)

97.94 / 90.02 %

SDSS / DESI benchmarks
(test split)

Code Map

Rules: literature first · accuracy and speed together · a version and changelog per merge

Get_data/

29 archive collectors
cache · pacing · mirror failover

Preprocess/

Stage 1–5
rule table · trained models (npz)

Classifier/

Subclass axes · branches
WD · variability · AGB …

app/

Qt app · sky map
bookmarks · packaging

benchmark/

Truth sets · scoring
independent-field scores

Docs

LITERATURE*.md (references)
WORK_REPORT_*.md · CHANGELOG

Branches: main (code · score tables) · validation-data (raw validation data, 233 MB) · v<version> tag → app release

Open Issues

Based on 208 errors among 24,663 independent-field objects

1 **Faint point-like galaxies**

At r > 21, PSF-fit galaxies carry no morphological signal → need evidence from other bands

galaxy → star 53

2 **Blended sources**

fracflux features cost −0.15 pp on independent fields → not adopted; needs a fix at association

26 % of errors

3 **Globular cluster · Nebula**

Cluster membership list (Vasiliev & Baumgardt 2021) not collected yet

star → unknown 40